$-R_1$

$$\sim \begin{bmatrix} 1 & 49 & -3 & : & 14 \\ 3 & 26 & 2 & : & 9 \\ 7 & 2 & 10 & : & 5 \end{bmatrix}$$

$R_2 - 3R_1$, $R_3 - 7R_1$

$$\sim \begin{bmatrix} 1 & 49 & -3 & : & 14 \\ 0 & -121 & 11 & : & -33 \\ 0 & -341 & 31 & : & -93 \end{bmatrix}$$

$1/11\ R_2$, $1/31\ R_3$

$$\sim \begin{bmatrix} 1 & 49 & -3 & : & 14 \\ 0 & \boxed{-11} & 1 & : & -3 \\ 0 & -11 & 1 & : & -3 \end{bmatrix}$$

$R_3 - R_2$

$$\sim \begin{bmatrix} 1 & 49 & -3 & : & 14 \\ 0 & -11 & 1 & : & -3 \\ 0 & 0 & 0 & : & 0 \end{bmatrix}$$

$\rho(A) = 2$ & $\rho(A:B) = 2$

$\rho(A) = \rho(A:B) = 2 < 3$ infinite sol.

$$A \quad X = B$$

$$\begin{bmatrix} 1 & 49 & -3 \\ 0 & -11 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 14 \\ -3 \\ 0 \end{bmatrix}$$

$$x + 49y - 3z = 14$$
$$-11y + z = -3$$

let $y = k$

$$z = -3 + 11k$$

$$x + 49k - 3(-3 + 11k) = 14$$
$$x + 49k + 9 - 33k = 14$$
$$\boxed{x = -16k + 5}$$

* For diff. values of k, system of eq^{n}s have diff. sol^{n}s

Consistency of Homogenous Equations

$$a_{11}x_1 + a_{12}x_2 + \dots + a_{1n}x_n = 0$$
$$a_{21}x_1 + a_{22}x_2 + \dots + a_{2n}x_n = 0$$
$$\vdots \qquad\qquad \vdots$$
$$a_{m1}x_1 + a_{m2}x_2 + \dots + a_{mn}x_n = 0$$

Matrix Form

$$\begin{bmatrix} a_{11} & a_{12} & \dots & a_{1n} \\ a_{21} & a_{22} & & a_{2n} \\ \vdots & \vdots & & \vdots \\ a_{m1} & a_{m2} & & a_{mn} \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix} = 0$$

$$A \quad X = 0$$

Homogenous eq^n are always consistent.
Either unique sol^n or infinite sol^n

$X = 0$ is always sol^n of Homogenous eq^n.

Procedure to solve homogenous equations

By applying row transformation we reduce A matrix into triangular matrix and find the rank of A matrix.

Case I :

If $\rho(A) = n$ (no. of unknown)
then system of homogenous eq^n have unique sol^n
i.e $X = 0$.
It is also known as trivial sol^n.

It is also known as trivial sol^n.

Case II :

If $\rho(A) < n$ (no. of unknown)
then system of eq^n have infinite sol^n or it is known as Non-Trival sol^n.

For non-trivial sol^n $|A| = 0$ (if A is square matrix)

Q. Solve the system of eq^n

$$x + y + z + w = 0$$
$$x + 3y + 2z + w = 0$$
$$2x + z - w = 0$$

$$A = \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & 3 & 2 & 1 \\ 2 & 0 & 1 & -1 \end{bmatrix}$$

$R_2 - R_1$, $R_3 - 2R_1$

$$\sim \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 1 & 0 \\ 0 & -2 & -1 & -3 \end{bmatrix}$$

$R_3 + R_2$

$$\sim \begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & -3 \end{bmatrix}$$

$$\rho(A) = 3 < 4$$

The system of eq^n have non trivial sol^n/infinite sol^n.

$$AX = 0$$

$$\begin{bmatrix} 1 & 1 & 1 & 1 \\ 0 & 2 & 1 & 0 \\ 0 & 0 & 0 & -3 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \\ w \end{bmatrix} = 0$$

$$x + y + z + w = 0$$
$$2y + z = 0$$
$$w = 0$$

let $y = k$

$$z = -2k$$

$$\cancel{x+y} \quad x + k - 2k + 0 = 0$$
$$x = k$$

Q. Find the value of k so that the eq^n

$$x + ky + 3z = 0$$
$$4x + 3y + kz = 0$$
$$2x + y + 2z = 0$$

have non trivial sol^n.

Sol. For non trivial sol^n $|A| = 0$

$$A = \begin{bmatrix} 1 & k & 3 \\ 4 & 3 & k \\ 2 & 1 & 2 \end{bmatrix}$$

$$|A| = \begin{vmatrix} 1 & k & 3 \\ 4 & 3 & k \\ 2 & 1 & 2 \end{vmatrix} = 0$$

$$1(6-k) - k(8-2k) + 3(4-6) = 0$$
$$k(2k-9) = 0$$

$\boxed{k = 9/2}$ or $\boxed{k = 0}$

Orthogonal matrix

$AA' = I$

Inverse of orthogonal matrix $A^{-1} = A'$

Q. Show that det of orthogonal matrix is ± 1.

Sol.

$$AA' = I$$
$$|AA'| = |I|$$
$$|A|\,|A'| = 1$$
$$|A|^2 = 1$$
$$|A| = \pm 1$$

Q. If A and B are orthogonal matrix then Prove that AB is also orthogonal.

Sol.

$$\left.\begin{aligned} AA' &= I \\ BB' &= I \end{aligned}\right\} \text{given}$$

To prove: $(AB)(AB)' = I$

Proof :

$$\begin{aligned}(AB)(AB)' &= ABB'A' \\ &= AIA' \\ &= AA' \\ &= I\end{aligned}$$

$\therefore$ AB is also orthogonal.

Q. Prove that the matrix is orthogonal

$$A = \begin{bmatrix} -2/3 & 1/3 & 2/3 \\ 2/3 & 2/3 & 1/3 \\ 1/3 & -2/3 & 2/3 \end{bmatrix}$$

Sol.

$$A' = \begin{bmatrix} -2/3 & 2/3 & 1/3 \\ 1/3 & 2/3 & -2/3 \\ 2/3 & 1/3 & 2/3 \end{bmatrix}$$

$$AA' = I$$

$$= \begin{bmatrix} -2/3 & 1/3 & 2/3 \\ 2/3 & 2/3 & 1/3 \\ 1/3 & -2/3 & 2/3 \end{bmatrix} \begin{bmatrix} -2/3 & 2/3 & 1/3 \\ 1/3 & 2/3 & -2/3 \\ 2/3 & 1/3 & 2/3 \end{bmatrix}$$

$$= \frac{1}{9} \begin{bmatrix} -2 & 1 & 2 \\ 2 & 2 & 1 \\ 1 & -2 & 2 \end{bmatrix} \begin{bmatrix} -2 & 2 & 1 \\ 1 & 2 & -2 \\ 2 & 1 & 2 \end{bmatrix}$$

$$= \frac{1}{9} \begin{bmatrix} (4+1+4) & (-4+2+2) & (-2-2+4) \\ (-4+2+2) & (4+4+1) & (2-4+2) \\ (-2-2+4) & (2-4+2) & (1+4+4) \end{bmatrix}$$

$$= \frac{1}{9} \begin{bmatrix} 9 & 0 & 0 \\ 0 & 9 & 0 \\ 0 & 0 & 9 \end{bmatrix}$$

$$= 9 \times \frac{1}{9} \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix}$$

$$= I$$

Hence proved!

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ASSIGNMENT 26/08/2019

Q1. Find the non singular matrix P & Q such that PAQ is in the normal form, also find rank of A.

$$A = \begin{bmatrix} 0 & 2 & 3 & 4 \\ 2 & 3 & 5 & 4 \\ 4 & 8 & 13 & 12 \end{bmatrix}$$

Q2. Find Inverse using Gauss-Jordan Method

$$A = \begin{bmatrix} 8 & 4 & 3 \\ 2 & 1 & 1 \\ 1 & 2 & 1 \end{bmatrix}$$

Q3.

$$\begin{aligned} 2x - y + z &= 4 \\ x + y + z &= 1 \\ x - 3y - 2z &= 2 \end{aligned}$$

Solve the eq^n (system of eq^n)

Assign

Soln 1.

$$A = \begin{bmatrix} 0 & 2 & 3 & 4 \\ 2 & 3 & 5 & 4 \\ 4 & 8 & 13 & 12 \end{bmatrix}$$

$$A_{3\times4} = I_{3\times3}\, A\, I_{4\times4}$$

$$\begin{bmatrix} 0 & 2 & 3 & 4 \\ 2 & 3 & 5 & 4 \\ 4 & 8 & 13 & 12 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$R_1 \rightarrow R_2$

$$\begin{bmatrix} 2 & 3 & 5 & 4 \\ 0 & 2 & 3 & 4 \\ 4 & 8 & 13 & 12 \end{bmatrix} = \begin{bmatrix} 0 & 1 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$R_1 \rightarrow R_1/2$

$$\begin{bmatrix} 1 & 3/2 & 5/2 & 2 \\ 0 & 2 & 3 & 4 \\ 4 & 8 & 13 & 12 \end{bmatrix} = \begin{bmatrix} 0 & 1/2 & 0 \\ 1 & 0 & 0 \\ 0 & 0 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$R_3 - 4R_1$

$$\begin{bmatrix} 1 & 3/2 & 5/2 & 2 \\ 0 & 2 & 3 & 4 \\ 0 & 2 & 3 & 4 \end{bmatrix} = \begin{bmatrix} 0 & 1/2 & 0 \\ 1 & 0 & 0 \\ 0 & -2 & 1 \end{bmatrix} A \begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$C_2 - 3/2\, C_1$ $\quad$ $C_3 - 5/2\, C_1$ $\quad$ $C_4 - 2C_1$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 3 & 4 \\ 0 & 2 & 3 & 4 \end{bmatrix} = \begin{bmatrix} 0 & 1/2 & 0 \\ 1 & 0 & 0 \\ 0 & -2 & 1 \end{bmatrix} A \begin{bmatrix} 1 & -3/2 & -5/2 & -2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$R_3 - R_2$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 2 & 3 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1/2 & 0 \\ 1 & 0 & 0 \\ -1 & -2 & 1 \end{bmatrix} A \begin{bmatrix} 1 & -3/2 & -5/2 & -2 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$C_2/2$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 3 & 4 \\ 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1/2 & 0 \\ 1 & 0 & 0 \\ -1 & -2 & 1 \end{bmatrix} A \begin{bmatrix} 1 & -3/4 & -5/2 & -2 \\ 0 & 1/2 & 0 & 0 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$C_3 - 3C_2$ $\quad$ $C_4 - 4C_2$

$$\begin{bmatrix} 1 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 \end{bmatrix} = \begin{bmatrix} 0 & 1/2 & 0 \\ 1 & 0 & 0 \\ -1 & -2 & 1 \end{bmatrix} A \begin{bmatrix} 1 & -3/4 & -1/4 & 1 \\ 0 & 1/2 & -3/2 & -2 \\ 0 & 0 & 1 & 0 \\ 0 & 0 & 0 & 1 \end{bmatrix}$$

$$\begin{bmatrix} I_2 & 0 & 0 \\ 0 & 0 & 0 \end{bmatrix} = PAQ$$

$$\rho(A) = 2$$

Assign

Sol2.

$$A = \begin{bmatrix} 8 & 4 & 3 \\ 2 & 1 & 1 \\ 1 & 2 & 1 \end{bmatrix}$$

$$A = I A$$

$$\begin{bmatrix} 8 & 4 & 3 \\ 2 & 1 & 1 \\ 1 & 2 & 1 \end{bmatrix} = \begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 1 \end{bmatrix} A$$

$$R_{13}$$

$$\begin{bmatrix} 1 & 2 & 1 \\ 2 & 1 & 1 \\ 8 & 4 & 3 \end{bmatrix} = \begin{bmatrix} 0 & 0 & 1 \\ 0 & 1 & 0 \\ 1 & 0 & 0 \end{bmatrix} A$$

$$R_2 - 2R_1 \quad , \quad R_3 - 8R_1$$

$$[A:I] = \left[\begin{array}{ccc:ccc} 1 & 2 & 1 & 0 & 0 & 1 \\ 0 & -3 & -1 & 0 & 1 & -2 \\ 0 & -12 & -5 & 1 & 0 & -8 \end{array}\right]$$

$$-R_2/3$$

$$\left[\begin{array}{ccc:ccc} 1 & 2 & 1 & 0 & 0 & 1 \\ 0 & 1 & 1/3 & 0 & -1/3 & 2/3 \\ 0 & -12 & -5 & 1 & 0 & -8 \end{array}\right]$$

$$R_1 - 2R_2 \qquad R_3 + 12R_2$$

$$\left[\begin{array}{ccc:ccc} 1 & 0 & 1/3 & 0 & 2/3 & -1/3 \\ 0 & 1 & 1/3 & 0 & -1/3 & 2/3 \\ 0 & 0 & -1 & 1 & -4 & 0 \end{array}\right]$$

$$R_1 + 1/3 R_3 \qquad R_2 + 1/3 R_3$$

$$\left[\begin{array}{ccc:ccc} 1 & 0 & 0 & 1/3 & -2/3 & -1/3 \\ 0 & 1 & 0 & 1/3 & -5/3 & 2/3 \\ 0 & 0 & 1 & 1 & -4 & 0 \end{array}\right]$$

Assign

Sol^n (3)

$$2x - y + z = 4$$
$$x + y + z = 1$$
$$x - 2y - 2z = 2$$

$$AX = B$$

$$\begin{bmatrix} 2 & -1 & 1 \\ 1 & 1 & 1 \\ 1 & -2 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix}$$

$$[A:B] = \left[\begin{array}{ccc:c} 2 & -1 & 1 & 4 \\ 1 & 1 & 1 & 1 \\ 1 & -3 & -2 & 2 \end{array}\right]$$

$$= \left[\begin{array}{ccc:c} 1 & 1 & 1 & 1 \\ 2 & -1 & 1 & 4 \\ 1 & -2 & -2 & 2 \end{array}\right]$$

$$\begin{array}{cc} R_2 - 2R_1 & R_3 - R_1 \end{array}$$

$$= \left[\begin{array}{ccc:c} 1 & 1 & 1 & 1 \\ 0 & -3 & -1 & 2 \\ 0 & -3 & -3 & 1 \end{array}\right]$$

$$R_3 - R_2$$

$$= \left[\begin{array}{ccc:c} 1 & 1 & 1 & 1 \\ 0 & -3 & -1 & 2 \\ 0 & 0 & -2 & -1 \end{array}\right]$$

$\rho(A) = 3$

$\rho(A:B) = 3$

$\rho(A) = \rho(A:B) = 3 = n$

$\therefore$ The system of equations have unique sol^n

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$$AX = B$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & -3 & -1 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 2 \\ -1 \end{bmatrix}$$

$$x + y + z = 1$$

$$-3y - z = 2$$

$$-2z = -1$$

$$\boxed{z = \frac{1}{2}}$$

$$-3(y) - \frac{1}{2} = 2$$

$$-6y - 1 = 4$$

$$\boxed{y = \frac{-5}{6}}$$

$$x + \left(\frac{-5}{6}\right) + \left(\frac{1}{2}\right) = 1$$

$$\boxed{x = \frac{4}{3}}$$

27/08/19

Eigen values

Characteristic Equation

If A is any square matrix of order 'nxn' then form a matrix $[A - \lambda I]$ where λ is a scalar and I is a unit matrix

The det of $A - \lambda I = 0$ is known as characteristic eqn.

$$|A - \lambda I| = 0 \qquad \text{(Characteristic eq}^n\text{)}$$

Generalized Form

$$(-1)^n \lambda^n + K_1 \lambda^{n-1} + K_2 \lambda^{n-2} + \ldots + K_n = 0$$

The roots of characteristic eqn are known as Latent roots or Eigen values or characteristic roots.

Q. Find the Eigen values of a matrix

$$A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$$

Soln Characteristic eqn

$$|A - \lambda I| = 0$$

$$\begin{vmatrix} -2-\lambda & 2 & -3 \\ 2 & 1-\lambda & -6 \\ -1 & -2 & 0-\lambda \end{vmatrix} = 0$$

$$-(2+\lambda)\left[(1-\lambda)(-\lambda) - 12\right] - 2\left[-2\lambda - 6\right] - 3\left[-4 + 1 - \lambda\right]$$

$$-\lambda^3 - \lambda^2 + 21\lambda + 45 = 0 \rightarrow \text{characteristic Equation}$$

General formula

$$-\lambda^3 + (a_{11} + a_{22} + a_{33})\lambda^2 - (m_{11} + m_{22} + m_{33})\lambda + |A| = 0$$

by hit & trial $\lambda = -3$, $(\lambda + 3)$ is factor.

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Synthetic division

-3	-1	-1	21	45
	$-$	3	-6	-45
	-1	2	15	0
	λ^2	λ	c	

$$(\lambda+3)\ (-\lambda^2 + 2\lambda + 15) = 0$$
$$(\lambda+3)\ (\lambda^2 - 2\lambda - 15) = 0$$
$$(\lambda+3)\ (\lambda^2 - 5\lambda + 3\lambda - 15) = 0$$
$$(\lambda+3)\ (\lambda - 5)\ (\lambda+3) = 0$$
$$\boxed{\lambda = -3, -3, 5} \rightarrow \text{(Eigen values)}$$

[Sum of EV = Sum of diagonals]

Q. Find the characteristic equations & Eigen values of matrix

$$A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 5 & 1 \\ 3 & 1 & 1 \end{bmatrix}$$

Soln Characteristic Eq^n $|A - \lambda I| = 0$

$$\begin{vmatrix} 1-\lambda & 1 & 3 \\ 1 & 5-\lambda & 1 \\ 3 & 1 & 1-\lambda \end{vmatrix} = 0$$

$$1-\lambda\left[(5-\lambda)(1-\lambda) - 1\right] - \left[(1-\lambda) - 3\right] + 3\left[1 - (15 - 3\lambda)\right]$$

$$-\lambda^3 + 7\lambda^2 - (0)\lambda - 36 = 0$$
$$-\lambda^3 + 7\lambda^2 - 36 = 0 \quad \text{Characteristic Eq}^n$$

$$(\lambda+2)\ (-\lambda^2 + 9\lambda - 18) = 0$$
$$(\lambda+2)\ (\lambda^2 - 9\lambda + 18) = 0$$
$$(\lambda+2)\ (\lambda^2 - 6\lambda - 3\lambda + 18) = 0$$
$$(\lambda+2)\ (\lambda - 3)\ (\lambda - 6) = 0$$
$$\boxed{\lambda = -2, 3, 6} \quad \text{Eigen values}$$

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Q1. Find the Eigen values of matrix

$$A = \begin{bmatrix} -9 & 2 & 6 \\ 5 & 0 & -3 \\ -16 & 4 & 11 \end{bmatrix}$$

Q2. Solve the system of Eq^n

$$2x - y + z = 4$$
$$x + y + z = 1$$
$$x - 3y - 2z = 2$$

Assign ① Characteristic eq^n $|A - \lambda I| = 0$

$$\begin{vmatrix} -9-\lambda & 2 & 6 \\ 5 & 0-\lambda & -3 \\ -16 & 4 & 11-\lambda \end{vmatrix}$$

$$(-9-\lambda)\left[(-\lambda)(11-\lambda) + 12\right] - 2\left[5(11-\lambda) - 48\right] + 6\left[20 - 16\lambda\right]$$

$$\boxed{-\lambda^3 + 2\lambda^2 + \lambda - 2 = 0} \rightarrow \text{Characteristic eq}^n$$

$$(\lambda - 1)(-\lambda^2 + \lambda) = 0$$
$$(\lambda - 1)(\lambda^2 - \lambda) = 0$$
$$(\lambda - 1)\lambda(\lambda - 1) = 0$$

$$\boxed{\lambda = 1, 0, 1} \rightarrow \text{Eigen values}$$

1	-1	2	1	-2
		-1	1	2
	-1	1	2	0

$$(\lambda - 1)(-\lambda^2 + \lambda + 2) = 0$$
$$(\lambda - 1)(\lambda^2 - 2\lambda + \lambda - 2) = 0$$
$$(\lambda - 1)(\lambda + 1)(\lambda - 2) = 0$$

$$\boxed{\lambda = -1, 1, 2} \rightarrow \text{Eigen values}$$

Assign
Sol (2)

$$AX = B$$

$$\begin{bmatrix} 2 & -1 & 1 \\ 1 & 1 & 1 \\ 1 & -3 & -2 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 4 \\ 1 \\ 2 \end{bmatrix}$$

$$[A:B] = \left[\begin{array}{ccc:c} 2 & -1 & 1 & 4 \\ 1 & 1 & 1 & 1 \\ 1 & -3 & -2 & 2 \end{array}\right]$$

$$R_{12}$$

$$\left[\begin{array}{ccc:c} 1 & 1 & 1 & 1 \\ 2 & -1 & 1 & 4 \\ 1 & -3 & -2 & 2 \end{array}\right]$$

$$R_2 - 2R_1 \qquad R_3 - R_1$$

$$\left[\begin{array}{ccc:c} 1 & 1 & 1 & 1 \\ 0 & -3 & -1 & 2 \\ 0 & -4 & -3 & 1 \end{array}\right]$$

$$R_{23}$$

$$\left[\begin{array}{ccc:c} 1 & 1 & 1 & 1 \\ 0 & -4 & -3 & 1 \\ 0 & -3 & -1 & 2 \end{array}\right]$$

$$R_2 - R_3$$

$$\left[\begin{array}{ccc:c} 1 & 1 & 1 & 1 \\ 0 & -1 & -2 & -1 \\ 0 & -3 & -1 & 2 \end{array}\right]$$

$$-R_2$$

$$\left[\begin{array}{ccc:c} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & -3 & -1 & 2 \end{array}\right]$$

$$R_3 + 3R_2$$

$$\left[\begin{array}{ccc:c} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 5 & 5 \end{array}\right]$$

$1/5 \; R_3$

$$\left[\begin{array}{ccc:c} 1 & 1 & 1 & 1 \\ 0 & 1 & 2 & 1 \\ 0 & 0 & 1 & 1 \end{array}\right]$$

$\rho(A) = 3$

$\rho(A:B) = 3$

$\rho(A) = \rho(A:B) = 3 < 3$

The system of eq^n have unique sol^n

$AX = B$

$$\begin{bmatrix} 1 & 1 & 1 \\ 0 & 1 & 2 \\ 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} x \\ y \\ z \end{bmatrix} = \begin{bmatrix} 1 \\ 1 \\ 1 \end{bmatrix}$$

$$x + y + z = 1 \quad — (1)$$

$$y + 2z = 1 \quad — (2)$$

$$\boxed{z = 1} \quad — (3)$$

from (2) and (3)

$$y = 1 - 2$$

$$\boxed{y = -1} \quad — (4)$$

from (2) (3) and (4)

$$x + (-1) + (1) = 1$$

$$\boxed{x = 1}$$

28/08/19

Eigen vector

$$[A - \lambda I]\, X = 0$$

$$AX = \lambda X$$
$$AX - \lambda X = 0$$
$$[A - \lambda I]X = 0$$

where $X = \begin{bmatrix} x_1 \\ x_2 \\ \vdots \\ x_n \end{bmatrix}$

X is Eigen vector

Q Find the Eigen values and Eigen vector of the matrix

$$A = \begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix}$$

Sol. Characteristic eq^n $|A - \lambda I| = 0$

$$\begin{vmatrix} 8-\lambda & -6 & 2 \\ -6 & 7-\lambda & -4 \\ 2 & -4 & 3-\lambda \end{vmatrix} = 0$$

$$(8-\lambda)\left[(7-\lambda)(3-\lambda) - 16\right] + 6\left[-6(3-\lambda) + 8\right] + 2\left[24 - 14 + 2\lambda\right] = 0$$

$$\boxed{-\lambda^3 + 18\lambda^2 - 45\lambda = 0} \rightarrow \text{Characteristic eq}^n$$

$$-\lambda(\lambda^2 - 18\lambda - 45) = 0$$
$$\lambda(\lambda - 3)(\lambda - 15) = 0$$

$$\boxed{\lambda = 0, 3, 15} \rightarrow \text{Eigen values.}$$

NOTE : Apply cross multiplication only when eq^{n}s are independent.

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Eigen vector

$$[A-\lambda I]\,X = 0$$

$$\begin{bmatrix} 8-\lambda & -6 & 2 \\ -6 & 7-\lambda & -4 \\ 2 & -4 & 3-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

For $\lambda = 0$

$$\begin{bmatrix} 8 & -6 & 2 \\ -6 & 7 & -4 \\ 2 & -4 & 3 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$8x_1 - 6x_2 + 2x_3 = 0$$
$$-6x_1 + 7x_2 - 4x_3 = 0$$
$$2x_1 - 4x_2 + 3x_3 = 0$$

$$\frac{x_1}{24-14} = \frac{x_2}{-12+32} = \frac{x_3}{56-36}$$

$$\frac{x_1}{10} = \frac{x_2}{20} = \frac{x_3}{20}$$

$(1, 2, 2)$

For $\lambda = 3$

$$5x_1 - 6x_2 + 2x_3 = 0$$
$$-6x_1 + 4x_2 - 4x_3 = 0$$
$$2x - 4x_2 + 0 \cdot x_3 = 0$$

$$\frac{x_1}{24-8} = \frac{x_2}{-12+20} = \frac{x_3}{20-36}$$

$$\frac{x_1}{16} = \frac{x_2}{8} = \frac{x_3}{-10}$$

$(2, 1, -2)$

For $\lambda = 15$

$$-7x_1 - 6x_2 + 2x_3 = 0$$
$$-6x_1 - 8x_2 - 4x_3 = 0$$
$$2x_1 - 4x_2 - 12x_3 = 0$$

$$\frac{x_1}{24+16} = \frac{x_2}{-12-28} = \frac{x_3}{56-36}$$

$$\frac{x_1}{40} = \frac{x_2}{-40} = \frac{x_3}{20}$$

$(2, -2, 1)$

Q Find the Eigen values and Eigen vector of the matrix $A = \begin{bmatrix} 6 & -2 & 2 \\ -2 & 3 & -1 \\ 2 & -1 & 3 \end{bmatrix}$

Sol. Characteristic eq^n $|A - \lambda I| = 0$

$$\begin{vmatrix} 6-\lambda & -2 & 2 \\ -2 & 3-\lambda & -1 \\ 2 & -1 & 3-\lambda \end{vmatrix} = 0$$

$6(9-1) = 48$
$+2(-6+2) = -8$
$+2(2-6) = -8$
det $A = 32$

$-\lambda^3 + 12\lambda^2 - 36\lambda + 32 = 0 \rightarrow$ characteristic eq^n

$(\lambda - 2)(-\lambda^2 + 10\lambda - 16) = 0$

$(\lambda - 2)(\lambda^2 - 10\lambda + 16) = 0$

$(\lambda - 2)(\lambda - 2)(\lambda - 8) = 0$

$\boxed{\lambda = 2, 2, 8} \rightarrow$ Eigen values

$8 = 8$
$18 - 4 = 14$
$18 - 4 = 14$
minors of diagonal 36

NOTE: To find the Eigen vector of dependent eq^n choose any one variable = 0
Do not apply cross multiplication.

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Eigen vector:

I • For $\lambda = 2$

$$[A - \lambda I]\, X = 0$$

$$\begin{bmatrix} 6-\lambda & -2 & 2 \\ -2 & 3-\lambda & -1 \\ 2 & -1 & 3-\lambda \end{bmatrix} \begin{bmatrix} x_1 \\ y_1 \\ z_1 \end{bmatrix} = 0$$

$$4x_1 - 2y_1 + 2z_1 = 0$$
$$-2x_1 + y_1 - z_1 = 0$$
$$2x_1 - y_1 + z_1 = 0$$

Let $z_1 = 0$

$$4x_1 - 2y_1 = 0$$
$$4x_1 = 2y_1$$
$$\frac{x_1}{2} = \frac{y_1}{4}$$

$(1, 2, 0)$

II • Choose $x_1 = 0$ (as two values of λ are 2)

$$-2y_2 + 2z_2 = 0$$
$$-2y_2 = -2z_2$$

$$\frac{y_2}{1} = \frac{z_2}{1}$$

$(0, 1, -1)$

III. For $\lambda = 8$

$$-2x_3 - 2y_3 + 2z_3 = 0$$
$$-2x_3 - 5y_3 - z_3 = 0$$
$$2x_3 - y_3 - 5z_3 = 0$$

$$\frac{x_3}{2+10} = \frac{y_3}{-4-2} = \frac{z_1}{10-4}$$

$$(12, -6, 6)$$
$$(2, -1, 1)$$

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Q. Find the Eigen values and Eigen vector of the matrix

$$A = \begin{bmatrix} 2 & 1 & 1 \\ 1 & 2 & 1 \\ 0 & 0 & 1 \end{bmatrix}$$

Soln. Characteristic eq^n $|A - \lambda I| = 0$

$$\begin{vmatrix} 2-\lambda & 1 & 1 \\ 1 & 2-\lambda & 1 \\ 0 & 0 & 1-\lambda \end{vmatrix} = 0$$

1	-1	5	-7	3
		-1	4	-3
	-1	4	-3	0

$$-\lambda^3 + 5\lambda^2 - 7\lambda + 3 = 0$$

$$(\lambda-1)(-\lambda^2 + 4\lambda - 3) = 0$$
$$(\lambda-1)(\lambda^2 - 4\lambda + 3) = 0$$
$$(\lambda-1)(\lambda^2 - 3\lambda - \lambda + 3)$$
$$(\lambda-1)(\lambda-1)(\lambda-3)$$

Eigen values $(1, 1, 3)$

Eigen vector

- for $\lambda = 1$

$$AX = 0$$

$$\begin{bmatrix} 1 & 1 & 1 \\ 1 & 1 & 1 \\ 0 & 0 & 0 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$x_1 + x_2 + x_3 = 0$$

$$x_1 + x_2 + x_3 = 0$$

Now let $x_3 = 0$

$$x_1 + x_2 = 0$$

$$x_1 = -x_2$$

$$-x_1 = x_2$$

$$(-1, 1, 0)$$

- let $x_2 = 0$

$$x_1 + x_3 = 0$$

$$x_1 = -x_3$$

$$x_1 = x_3$$

$$(-1, 0, 1)$$

- for $\lambda = 3$

$$\begin{bmatrix} -1 & 1 & 1 \\ 1 & -1 & 1 \\ 0 & 0 & -2 \end{bmatrix} \begin{bmatrix} x_1 \\ x_2 \\ x_3 \end{bmatrix} = 0$$

$$-x_1 + x_2 + x_3 = 0$$

$$x_1 - x_2 + x_3 = 0$$

$$-2x_3 = 0$$

$$\frac{x_1}{-2} = \frac{x_2}{2} = \frac{x_3}{0}$$

$$(-1, 1, 0)$$

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Properties of Eigen values

1. The sum of Eigen values = Sum of principal diagonal elements

$$\lambda_1 + \lambda_2 + \lambda_3 = a_{11} + a_{22} + a_{33}$$

Proof: Let A be the square matrix of order 3×3 & $\lambda_1, \lambda_2, \lambda_3$ are Eigen values

$$A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$$

$$|A - \lambda I| = \begin{vmatrix} a_{11} - \lambda & a_{12} & a_{13} \\ a_{21} & a_{22} - \lambda & a_{23} \\ a_{31} & a_{32} & a_{33} - \lambda \end{vmatrix}$$

$$|A - \lambda I| = -\lambda^3 + (a_{11} + a_{22} + a_{32})\lambda^2 + \ldots\ldots \quad -(1)$$

If $\lambda_1, \lambda_2, \lambda_3$ are eigen values

$$|A - \lambda I| = -(\lambda - \lambda_1)(\lambda - \lambda_2)(\lambda - \lambda_3)$$

$$|A - \lambda I| = -\lambda^3 + (\lambda_1 + \lambda_2 + \lambda_3)\lambda^2 + \ldots\ldots \quad -(2)$$

from eq (1) & (2)

$$-\lambda^3 + (a_{11} + a_{22} + a_{33})\lambda^2 + \ldots = -\lambda^3 + (\lambda_1 + \lambda_2 + \lambda_3)\lambda^2 + \ldots$$

Compare coeff. of λ^2

$$\boxed{a_{11} + a_{22} + a_{33} = \lambda_1 + \lambda_2 + \lambda_3}$$

(2.) The Eigen values of a square matrix A and A' are same.

Proof: Let $A = \begin{bmatrix} a_{11} & a_{12} & a_{13} \\ a_{21} & a_{22} & a_{23} \\ a_{31} & a_{32} & a_{33} \end{bmatrix}$

$$|A - \lambda I| = \begin{vmatrix} a_{11}-\lambda & a_{12} & a_{13} \\ a_{21} & a_{22}-\lambda & a_{23} \\ a_{31} & a_{32} & a_{33}-\lambda \end{vmatrix}$$

$$|A' - \lambda I| = \begin{vmatrix} a_{11}-\lambda & a_{21} & a_{31} \\ a_{12} & a_{22}-\lambda & a_{32} \\ a_{13} & a_{23} & a_{33}-\lambda \end{vmatrix}$$

$\therefore$ $\boxed{|A - \lambda I| = |A' - \lambda I|}$ using property of det.

Characteristic eq^n of A & A' are same

$\therefore$ Eigen of A and A' are also same

(3.) The product of Eigen values = det of A matrix.

Proof: If $\lambda_1, \lambda_2 \ldots \lambda_n$ are eigen values of the matrix A, then the characteristic eq^n is

$$|A - \lambda I| = (-1)^n (\lambda - \lambda_1)(\lambda - \lambda_2) \ldots\ldots (\lambda - \lambda_n)$$

Put $\lambda = 0$

$$|A| = (-1)^n (-\lambda_1)(-\lambda_2) \ldots\ldots (-\lambda_n)$$
$$= (-1)^n (-1)^n \lambda_1 \lambda_2 \ldots\ldots \lambda_n$$

$\boxed{|A| = \lambda_1 \lambda_2 \lambda_3 \ldots\ldots \lambda_n}$ (n = order of matrix A)

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(4.) If λ is an eigen value of a non-singular matrix A then $\frac{1}{\lambda}$ is an eigen value of A^{-1}

Proof:

$$(A-\lambda I)X = 0$$

$$AX = \lambda X \qquad (\lambda \text{ is eigen value of } A)$$

Pre-multiply by A^{-1}

$$A^{-1}AX = A^{-1}(\lambda X)$$

$$IX = \lambda A^{-1}X$$

$$\frac{1}{\lambda}X = A^{-1}X$$

$$A^{-1}X = \frac{1}{\lambda}X$$

$\frac{1}{\lambda}$ is eigen value of A^{-1}

(5.) If λ is an eigen value of orthogonal matrix A then $\frac{1}{\lambda}$ is also its eigen value.

Proof: For orthogonal matrix

$$A' = A^{-1}$$

If λ is an eigen value of A, then $\frac{1}{\lambda}$ is eigen of A^{-1}

$\therefore$ $\frac{1}{\lambda}$ is also eigen of A^{-1}

The eigen values of A & A' are same

$\therefore$ $\frac{1}{\lambda}$ is also eigen of A.

Idempotent matrix :- $A^2 = A$

It is a matrix in which its square is equal to the matrix itself.

(6.) If $\lambda_1, \lambda_2, \ldots\ldots \lambda_n$ are the eigen values of A matrix then, $\lambda_1^m, \lambda_2^m, \ldots\ldots \lambda_n^m$ are the eigen values of A^m.

Proof : If λ is an eigen value of A then

$$AX = \lambda X$$

Pre multiply both sides by A

$$A^2 X = \lambda (AX)$$
$$A^2 X = \lambda^2 X$$

Again Pre multiply by A

$$A^3 X = \lambda^2 (AX)$$
$$A^3 X = \lambda^3 X$$

$$\boxed{A^m X = \lambda^m X}$$

From this relation it is clear, λ^m is an eigen value of A^m.

(7.) The Eigen values of Idempotent matrix are either 0 or 1.

Proof : If λ is an eigen value of A then,

$$AX = \lambda X$$

Pre-multiply by A

$$A^2 X = \lambda A X$$
$$A^2 X = \lambda (\lambda X)$$
$$A X = \lambda^2 X$$
$$\lambda X = \lambda^2 X$$

$$\Rightarrow \quad \lambda^2 \lambda - \lambda x = 0$$
$$(\lambda^2 - \lambda) x = 0$$

$$\lambda^2 - \lambda = 0$$
$$\lambda(\lambda - 1) = 0$$

$$\boxed{\lambda = 0, 1}$$

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Q Find the Eigen values and Eigen vector of the matrix.

Univer. Question Imp.

7.5M

$$A = \begin{bmatrix} -2 & 2 & -3 \\ 2 & 1 & -6 \\ -1 & -2 & 0 \end{bmatrix}$$

Q Find the inverse

7.5M

$$A = \begin{bmatrix} 2 & 1 & -1 \\ 0 & 2 & 1 \\ 5 & 2 & -3 \end{bmatrix} \text{ OR } A = \begin{bmatrix} 1 & 1 & 3 \\ 1 & 3 & -3 \\ -2 & -4 & -4 \end{bmatrix}$$

Q Find the ~~inverse~~ rank of the matrix

$$A = \begin{bmatrix} 2 & 5 & 6 \\ 1 & 2 & 3 \\ 3 & 7 & 9 \end{bmatrix}$$

by both methods
① Minor ② Normal

Q Find the non-singular matrices P & Q such that PAQ is in the normal form

$$A = \begin{bmatrix} 3 & 2 & 1 \\ 2 & 1 & 1 \\ 1 & 2 & 3 \\ 3 & 2 & 2 \end{bmatrix}$$

Q2.

$$[A:I] = \left[\begin{array}{ccc:ccc} 1 & 1 & 3 & 1 & 0 & 0 \\ 1 & 3 & -3 & 0 & 1 & 0 \\ -2 & -4 & -4 & 0 & 0 & 1 \end{array}\right]$$

$-R_3, R_{13}$

$$\left[\begin{array}{ccc:ccc} 2 & 4 & 4 & 0 & 0 & -1 \\ 1 & 3 & -3 & 0 & 1 & 0 \\ 1 & 1 & 3 & 1 & 0 & 0 \end{array}\right]$$

$R_1 - R_2, \; R_2 - R_3$

$$\left[\begin{array}{ccc:ccc} 1 & 1 & 7 & 0 & -1 & -1 \\ 0 & 2 & -6 & -1 & 1 & 0 \\ 1 & 1 & 3 & 1 & 0 & 0 \end{array}\right]$$

$R_3 - R_1$

$$\left[\begin{array}{ccc:ccc} 1 & 1 & 7 & 0 & -1 & -1 \\ 0 & 2 & -6 & -1 & 1 & 0 \\ 0 & 0 & -4 & 0 & 1 & 1 \end{array}\right]$$

$\frac{1}{2} R_2$

$$\left[\begin{array}{ccc:ccc} 1 & 1 & 7 & 0 & -1 & -1 \\ 0 & 1 & -3 & -1 & 1 & 0 \\ 0 & 0 & -4 & 1 & 0 & 0 \end{array}\right]$$

$R_3 - R_1$

$$\left[\begin{array}{ccc:ccc} 1 & 1 & 7 & 0 & -1 & -1 \\ 0 & 2 & -6 & -1 & 1 & 0 \\ 0 & 0 & -4 & 0 & 1 & 1 \end{array}\right]$$

$\frac{1}{2} R_2$

$$\left[\begin{array}{ccc:ccc} 1 & 1 & 7 & 0 & -1 & -1 \\ 0 & 1 & -3 & -\frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & -4 & 0 & 1 & 1 \end{array}\right]$$

$R_1 - R_2$

$$\left[\begin{array}{ccc:ccc} 1 & 0 & 10 & -\frac{1}{2} & -\frac{3}{2} & -1 \\ 0 & 1 & -3 & -\frac{1}{2} & \frac{1}{2} & 0 \\ 0 & 0 & -4 & 0 & 1 & 1 \end{array}\right]$$

$-\frac{1}{4} R_3$

$$\left[\begin{array}{ccc:ccc} 1 & 0 & 10 & -\frac{1}{2} & -\frac{3}{2} & -1 \\ 0 & 1 & -3 & -\frac{1}{2} & -\frac{1}{2} & 0 \\ 0 & 0 & 1 & 0 & -\frac{1}{4} & -\frac{1}{4} \end{array}\right]$$

$R_2 + 3R_3$, $R_1 - 10R_3$

$$\left[\begin{array}{ccc:c} 1 & 0 & 0 & \\ 0 & 1 & 0 & \\ 0 & 0 & 1 & \end{array}\right] \left[\begin{array}{ccc} -\frac{1}{2} & 1 & \frac{3}{2} \\ -\frac{1}{2} & -\frac{5}{4} & -\frac{3}{4} \\ 0 & -\frac{1}{4} & -\frac{1}{4} \end{array}\right]$$

Q3.

① By minor method

$$|A| = \begin{vmatrix} 2 & 5 & 6 \\ 1 & 2 & 3 \\ 3 & 7 & 9 \end{vmatrix} = [2(18-21)] - 5(9-9) + 6(7-6)$$

$$= 0$$

Now det of minor $\begin{vmatrix} 2 & 5 \\ 1 & 2 \end{vmatrix} = 4 - 5$

$$= -1$$

Rank of matrix 2

or $\rho(A) = 2$

② By Normal method

$$\begin{bmatrix} 2 & 5 & 6 \\ 1 & 2 & 3 \\ 3 & 7 & 9 \end{bmatrix}$$

R_{12}

$$\begin{bmatrix} 1 & 2 & 3 \\ 2 & 5 & 6 \\ 3 & 7 & 9 \end{bmatrix}$$

$R_2 - 2R_1$, $R_3 - 3R_1$

$$\begin{bmatrix} 1 & 2 & 3 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$C_2 - 2C_1$, $C_3 - 3C_1$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 1 & 0 \end{bmatrix}$$

$R_3 - R_2$

$$\begin{bmatrix} 1 & 0 & 0 \\ 0 & 1 & 0 \\ 0 & 0 & 0 \end{bmatrix}$$

$$\begin{bmatrix} I_2 & 0 \\ 0 & 0 \end{bmatrix} \rightarrow \text{Normal form.}$$

$\rho(A) = 2$